

Research of Computer Forensics Based on Benford's Law

Shan Meijing

Institute of Information science and Technology, East China University of Political and Law,
Shanghai, China. 201620
mjshan47@gmail.com

Keywords: Benford's Law, Computer forensics, File analysis

Abstract: In this paper we construct model in computer forensics by using Benford's Law, which is one of three famous laws in statistics in linguistics. Especially, we use Benford's Law to analyze data in file analysis and Anomaly detection. We illustrate that Benford's Law is able to supply the strategy for data analysis. The future work, Benford's Law will be the theoretical rule for intelligent compression of mass data.

Introduction

Cybercrime is one of the fastest growing and hot areas of crime. More and more criminals are exploiting the speed, convenience and anonymity that modern technologies offer in order to commit a diverse range of criminal activities. How to attempt to capture as much as possible digital evidence of the entire cybercrime investigative process, how to efficiently and effectively combat cybercrime, are very urgent issues. Computer forensics is a branch of digital forensic science pertaining to legal evidence found in computers and digital storage media. The goal of computer forensics is to examine digital media in a forensically sound manner with the aim of identifying, preserving, recovering, analyzing and presenting facts and opinions about the information.[1]

For the particularity of the electronic evidence in computer forensics, computer forensics should comply with the corresponding discipline and principle. It may involve the acquisition and analysis of digital evidence, authentication of documents, identification of sources and suspects, and so forth. From 1990's, many proposed forensic models have been studied [2-5]. Pollitt [2] identified four steps in digital forensics, as follows: acquisition, identification, evaluation, and admission of evidence, and further described admissibility of evidence. Authors in [4] use XML and OEM data model, data fusion technology, forensics knowledge base, experts inference mechanism and mining engine to set up forensics model. Authors in [5] add time constraint and supervision procedure.

During the process of computer forensics, we need to analyze the data by knowledge of probability theory and statistics. In this paper, we construct the model in cyber forensics by Benford's law which is one of three laws in language statistics. We also use this law to analyze the file system and apply process anomaly detection. The rest content of the paper is indicated as follows: Section 2 mainly introduce the contents of Benford law and the related technologies in computer forensics. Section 3 studies the application of Benford law in computer forensics. Section 4 draw a conclusion.

Benford law

In 1881, Canadian-American astronomer and mathematician Simon Newcomb noticed that the beginning pages of the logarithmic table books were dirtier than the end pages, which indicated that the tables of logarithms were not used in a uniform way. [7]

Frank Benford was a research physicist at General Electric. In 1938, he also noticed the similar phenomenon. He found something unusual about a book of logarithmic tables. The first pages showed more wear than the last pages, indicating that numbers beginning with the digit 1 were being looked up more often than numbers beginning with 2 through 9. He published his result which about the frequency distribution of digits in many (but not all) real-life sources of data. In this distribution, the number 1 occurs as the leading digit about 30% of the time, while larger numbers occur in that position less frequently: 9 as the first digit less than 5% of the time. Benford's Law also concerns the expected distribution for digits beyond the first, which approach a uniform distribution.

Benford seized upon this idea and spent years collecting data to show that this pattern was widespread in nature. In 1938, Benford published his results, citing more than 20,000 values such as atomic weights, numbers in magazine articles, baseball statistics, and the areas of rivers.

Benford's Law(also called the First-Digit Law): A set of numbers is said to satisfy Benford's Law if the leading digit d ($d \in \{1, \dots, 9\}$) occurs with probability

$$P(d) = \log_{10}(d+1) - \log_{10}(d) = \log_{10}\left(\frac{d+1}{d}\right) = \log_{10}\left(1 + \frac{1}{d}\right) \quad (1)$$

The quantity $P(d)$ is proportional to the space between d and $d + 1$ on a logarithmic scale.

Nigrini appears to be the first researcher to apply Benford's law extensively to accounting numbers with the goal to detect fraud.[Nigrini 1996] More recently, papers have been published which detail practical applications of digital analysis such as descriptions of how an auditor performs tests on sets of accounting numbers, how an auditor uses digital analysis computer programs, and case studies for training students. However, Benford law does not suit for all the data, such as Lotto number, telephone number etc.

Table1 Based on the frequency of the digital distribution of Benford's law

	First digit	Second digit	Third digit	Fourth digit
0		.11968	.10178	.10018
1	.30103	.11389	.10138	.10014
2	.17609	.10882	.10097	.10010
3	.12494	.10433	.10057	.10006
4	.09691	.10031	.10081	.10002
5	.07918	.09668	.09979	.09998
6	.06695	.09337	.09940	.09994
7	.05799	.09035	.09902	.09990
8	.05119	.08757	.09864	.09986
9	.04576	.08500	.09827	.09982

Application of Benford law in Computer Forensics

Many evidences are potential in computer crime. The files the users set up may include important information

of crime activity. In many cases, suspects hide or modify the files, such as hide the important files in some partition or in some seems irrelevant file folders. Computer forensics should consider the properties of files. The tracked operations include time of files or file folders' creation, last modification, last access, movement and deletion, the owner, the access privilege.

When delete a file, the intuitive result is that the file name disappears from the directory list, and the file name and inode number of directory entries are marked unused. Some file systems, such as Microsoft FAT16 and FAT32, will hide the file name by a special configuration, then the file is marked for deletion. The traditional Berkeley Fast File System (FFS) will be disconnected directory entries, all file attributes and file data connections between data blocks. Extended FFS usually be found between Solaris and BSD system. Here we only discuss the cases which can access local file system, not about remote file system.

Actually, the files in computer system are not only the ones user set up but also the ones of operating system, which includes backup files, log files, configuration files, swap files, system files, hidden files, and temporary files etc. The sizes of these files obey the probabilistic distribution described in Benford Law. We got all the files in system disc by some instructions, and find the size of these files obey Benford law, shown in Table 2. In this table, "FSD" which starts with digit 1~9 means the size of file, "Total" means the number of files, "Actual" means the proportion of the file, "Benford" means the proportion of the file calculated by Benford law.

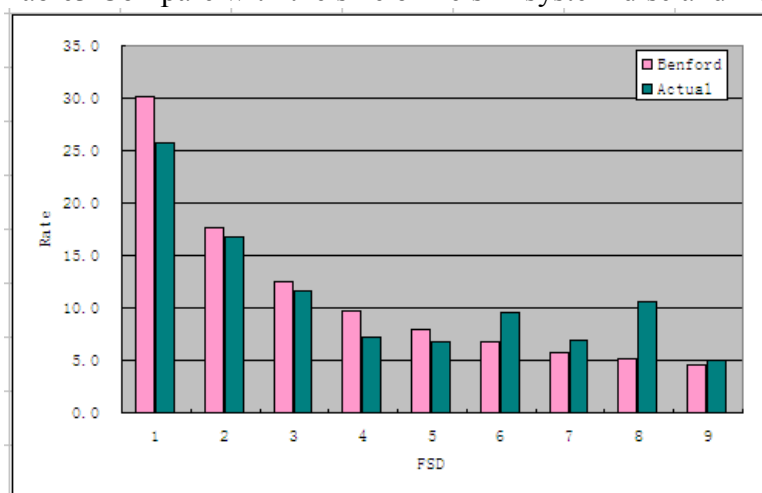
Table2 The size of file in system disk

FSD	Total	Actual(%)	Benford(%)
1	4451	25.7	30.1
2	2900	16.7	17.6
3	2003	11.6	12.5
4	1243	7.2	9.7
5	1181	6.8	7.9
6	1651	9.5	6.7
7	1207	7.0	5.8
8	1825	10.5	5.1
9	870	5.0	4.6
0	19		

Compare the data in “Actual” and “Benford” column in Table2, we find they are similar. We use 2 for example. There are 2900 files which start with 2 in system disk, accounting for 16.7% of the total file. The calculating result is 17.6%. We compare the results and give the compared analysis chart as Table3.

By a large mount of experiments, we find the size of file in computer has the similar law. It supplies theoretical judgment strategy for anomaly detection of files in computer forensics.

Table3 Compare with the size of fiels in system disc and Benford law



Conclusion

The research work of this paper is very rough. The further research work will use Benford law and formal method to study the digital evidences and models in computer forensics. We also plan to use knowledge of probability statistics to reduce the size of mass data with a TB level, for improving the analyzable of digital evidences.

Acknowledgements

This work was financially supported by Innovation Program of Shanghai Municipal Education Commission(No.12YS107) and East China University of Political and Law Scientific Research Project(No.12HZK028).

Reference

- [1] Michael G. Noblett; Mark M. Pollitt, Lawrence A. Presley (October 2000). Recovering and examining computer forensic evidence. Retrieved 26 July 2010.
- [2] M. Pollitt, Computer forensics: an approach to evidence in cyberspace, in Proceedings of the National Information Systems Security Conference, Baltimore, MD, 1995, vol. 2, pp. 487-491, <http://www.digitalevidencepro.com/Resources/Approach.pdf>.
- [3] Sabah Al-Fedaghi and Bashayer Al-Babtain. Modeling the Forensics Process. http://www.sersc.org/journals/IJSIA/vol6_no4_2012/9.pdf

-
- [4] Chen Long, Mai Yonghao. Computer Forensics Technology. Wuhan: Wuhan University Press. 2007.
- [5] Ding Liping, Wang Yongji. Multi-dimension Forensics Model. Computer Security-proceeding of 20th National Symposium on computer security. 2006: 5-9.
- [6] Newcomb S. Note on the frequency of use of the different digits in natural numbers. American Journal of Mathematics, 1881,4(1/4):39-40.
- [7] Benford F. The law of anomalous numbers. Proc of the American Philosophical. 1938:551-572.
- [8] Mark Nigrini, Joseph T. Wells. Benford's Law: Applications for Forensic Accounting, Auditing, and Fraud Detection. 2012.

Materials Science, Computer and Information Technology

10.4028/www.scientific.net/AMR.989-994

Research of Computer Forensics Based on Benford's Law

10.4028/www.scientific.net/AMR.989-994.2148